

Computer Networks

Complete Notes

Comprehensive short notes covering all topics

Source: GeeksforGeeks CN Tutorial

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1. Fundamentals of Computer Networks

1.1 What is a Computer Network?

A computer network is a group of two or more connected devices that communicate with each other to share data, resources, and services. Connections can be established using wired media (Ethernet, fiber) or wireless technologies (Wi-Fi, Bluetooth).

Basic Terminologies

- **Network:** A group of connected computers and devices that can communicate and share data.
- **Node:** Any device that can send, receive, or forward data (laptops, printers, servers, etc.).
- **Link:** The medium that connects nodes – cables, wires, or wireless signals.
- **Protocol:** A set of rules governing how data is sent and received. Each device is identified by a unique IP Address.
- **Firewall:** A security tool (hardware/software) that monitors traffic and allows, blocks, or drops data based on set rules.

Goals of Computer Networks

- **Resource Sharing:** Share printers, scanners, files across devices.
- **Communication:** Email, chat, video conferencing.
- **Remote Access:** Access data and systems from anywhere.
- **Reliability & Fault Tolerance:** Ensure system runs smoothly and recovers from failures.
- **Scalability:** Support growth in users, processes, and resources.

1.2 Types of Computer Networks

By Geographical Area

- **PAN (Personal Area Network):** 1–10 m range. Bluetooth, NFC, Infrared. E.g., smartphone to earbuds.
- **LAN (Local Area Network):** One room/building/campus. Ethernet, Wi-Fi. E.g., home or office Wi-Fi.
- **CAN (Campus Area Network):** Multiple LANs within a campus. Ethernet, Fiber Optics.
- **MAN (Metropolitan Area Network):** City-wide. Fiber Optics, Microwave, Metro Ethernet. E.g., city ISP network.
- **WAN (Wide Area Network):** Country/continent/global. Leased lines, Satellite, Internet. E.g., the Internet.

By Transmission Medium

- **Broadcast Network:** Shared medium, one-to-many. Data sent to all nodes; only intended device processes it. E.g., Ethernet (bus), Wi-Fi.
- **Point-to-Point Network:** Dedicated links, one-to-one. Data may pass through routers. E.g., leased lines, Internet.

By Ownership & Access

- **Private Network:** Restricted to authorized users. Higher security. E.g., corporate intranet, office LAN.
- **Public Network:** Open to general public. E.g., Internet, public Wi-Fi, mobile cellular networks.
- **Hybrid Network:** Combination of private and public. E.g., corporate network + Internet, online banking.

Internetwork: Intranet vs Extranet

- **Intranet:** Private network within an organization. Uses TCP/IP, HTTP, FTP. E.g., employee portals, internal emails.
- **Extranet:** Extension of intranet allowing limited access to external users (partners, vendors). Uses VPN/secure web. E.g., supplier management systems.

1.3 Network Architecture

- **Client-Server:** Server manages client nodes. Centralized resources and management.
- **Peer-to-Peer (P2P):** No central server. Each device can act as client or server.

1.4 Network Devices

Layer 1 – Physical Layer Devices

- **Hub:** Central connection point. Broadcasts to ALL ports. Obsolete – replaced by switches.
- **Repeater:** Regenerates weak signals to extend network range beyond 100m.
- **Modem:** Converts digital signals to analog (modulation) and vice versa (demodulation). Connects home to ISP.

Layer 2 – Data Link Layer Devices

- **Switch:** Learns MAC addresses, sends data ONLY to destination port. Reduces collisions. Types: Unmanaged (plug-and-play), Managed (VLANs, QoS).
- **Bridge:** Connects two LAN segments. Filters traffic based on MAC addresses. Largely replaced by switches.
- **Access Point (AP):** Creates WLAN. Acts as bridge between wired Ethernet and wireless Wi-Fi devices.

Layer 3 – Network Layer Devices

- **Router:** Connects multiple networks (e.g., home LAN to Internet). Uses IP addresses & routing tables for best path. Stops broadcast traffic.

- Router: Hybrid – routes known protocols (TCP/IP), bridges unknown protocols. Rarely used today.

Layer 4–7 – Advanced Devices

- Gateway: Translator between dissimilar networks. Converts protocols, data formats, architectures.
- Firewall: Controls incoming/outgoing traffic based on security rules. Packet filtering & stateful inspection.

Feature	Hub	Switch	Router
OSI Layer	Layer 1 (Physical)	Layer 2 (Data Link)	Layer 3 (Network)
Address Type	None (Bits)	MAC Address	IP Address
Data Flow	Broadcast (to all)	Unicast (to destination)	Route (best path)
Intelligence	Dumb	Smart	Smarter
Used For	Obsolete	Connecting Devices (LAN)	Connecting Networks (WAN)

2. Network Models (OSI & TCP/IP)

2.1 OSI Model (Open Systems Interconnection)

The OSI Model is a 7-layer reference framework developed by ISO that explains how different computer systems communicate over a network. Each layer has specific functions and responsibilities.

Layer 1: Physical Layer

- Transmits raw bits (0s and 1s) over physical medium (wires, fiber, wireless).
- Functions: Bit synchronization (clock), bit rate control, physical topologies, transmission mode (simplex, half-duplex, full-duplex).
- Devices: Hub, Repeater, Modem, Cables.
- Protocols: Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15.1), USB.

Layer 2: Data Link Layer (DLL)

- Responsible for node-to-node delivery. Data unit = Frame.
- Sublayers: LLC (Logical Link Control) – flow control, error messages; MAC (Media Access Control) – physical addressing, channel access.
- Functions: Framing, Physical addressing (MAC), Error control, Flow control, Access control.
- Devices: Switch, Bridge, NIC.
- Protocols: Ethernet, PPP, HDLC, SDLC.

Layer 3: Network Layer

- Handles logical addressing (IP) and routing. Data unit = Packet.
- Functions: Routing (best path selection), Logical addressing, Packetization, Fragmentation & reassembly, Subnetting, NAT.
- Devices: Router.
- Protocols: IP (IPv4/IPv6), ICMP, ARP, RARP, OSPF, BGP, MPLS.

Layer 4: Transport Layer

- End-to-end delivery. Data unit = Segment (TCP) / Datagram (UDP).
- Functions: Segmentation & reassembly, Service point addressing (port numbers), Error detection & recovery, Flow control, Multiplexing/demultiplexing.
- Protocols: TCP (reliable, connection-oriented), UDP (fast, connectionless), SCTP.

Layer 5: Session Layer

- Establishes, manages, and terminates sessions between two devices.
- Functions: Session establishment/maintenance/termination, Synchronization (checkpoints), Dialog control (half-duplex/full-duplex).
- Protocols: NetBIOS, RPC, PPTP.

Layer 6: Presentation Layer

- Translation layer – data formatting and manipulation.
- Functions: Translation (ASCII to EBCDIC), Encryption/decryption, Compression.
- Protocols/Standards: TLS/SSL, JPEG, MPEG, GIF, MIME.

Layer 7: Application Layer

- Closest to end user. Provides network services to applications.
- Functions: Network Virtual Terminal (NVT), File Transfer (FTAM), Mail services, Directory services.
- Protocols: HTTP, FTP, SMTP, DNS, DHCP, SNMP, Telnet, SSH.

Layer	Name	Data Unit	Key Protocols
7	Application	Data	HTTP, FTP, SMTP, DNS
6	Presentation	Data	TLS/SSL, JPEG, MIME
5	Session	Data	NetBIOS, RPC, PPTP
4	Transport	Segment/Datagram	TCP, UDP, SCTP
3	Network	Packet	IP, ICMP, ARP, OSPF
2	Data Link	Frame	Ethernet, PPP, HDLC
1	Physical	Bit	Ethernet, Wi-Fi, USB

Data flow: Application creates data → Presentation formats/encrypts → Session manages connection → Transport segments → Network packets & routes → Data Link frames → Physical transmits bits. Reversed at receiver.

2.2 TCP/IP Model

The TCP/IP model is a practical 4-layer networking framework used in the real Internet. Simpler and more widely implemented than OSI. Standardized by RFC 1122.

Layer 1: Network Access (Link) Layer

- Combines OSI Physical + Data Link layers. Handles physical transmission, framing, error detection, MAC addressing, access control.
- Sends/receives raw bits over Ethernet, fiber, Wi-Fi.

Layer 2: Internet Layer

- Equivalent to OSI Network layer. Logical addressing (IP), packet routing, fragmentation & reassembly.
- Protocols: IP (IPv4/IPv6), ICMP (error reporting, ping), ARP (IP to MAC mapping).

Layer 3: Transport Layer

- Same as OSI Transport layer. End-to-end delivery with segmentation, flow control, error handling.
- TCP: Connection-oriented, reliable, error-checking, retransmission, ordered delivery. Used for web, email, file transfer.
- UDP: Connectionless, no error-checking, no retransmission, faster. Used for streaming, gaming, VoIP.

Layer 4: Application Layer

- Combines OSI Session + Presentation + Application. User-facing. Supports HTTP, FTP, SMTP, DNS, etc.
- Handles data formatting, encryption, session tracking, and application-level protocols.

TCP/IP Layer	OSI Equivalent	Key Protocols
Application	Application + Presentation + Session	HTTP, FTP, SMTP, DNS, SSH
Transport	Transport	TCP, UDP, SCTP
Internet	Network	IP, ICMP, ARP, IGMP
Network Access	Data Link + Physical	Ethernet, Wi-Fi, PPP

Why TCP/IP over OSI?

- Simpler (4 layers vs 7). Protocol-driven (practical) vs theoretical. Open standard, flexible, robust.
- OSI is great for learning; TCP/IP is what is actually used in real-world networking.

3. Physical Layer

3.1 Functions

- Transmits raw data bits through physical hardware (cables, connectors, wireless channels).
- Encoding/Decoding: Converts data into signals for transmission and back.
- Modulation/Demodulation: Prepares data for transmission medium.
- Transmission Modes: Simplex (one-way), Half-duplex (two-way, one at a time), Full-duplex (simultaneous two-way).
- Bit synchronization, bit rate control.

3.2 Physical Topologies

Physical topologies describe the arrangement of devices and cables in a network.

Line Configurations

- Point-to-Point: Dedicated link between two devices.
- Multi-Point: Shared link connecting multiple devices.

Common Topologies

- Bus: All devices share a single backbone cable. Simple but single point of failure.
- Star: All devices connect to a central hub/switch. Easy to manage. Hub failure = network down.
- Ring: Each device connects to two neighbors forming a circle. Data travels in one direction. One break = network down.
- Mesh: Every device connects to every other device. Full mesh = $n(n-1)/2$ links. Most reliable but expensive.
- Tree (Hierarchical): Combination of star and bus. Root node, branches. Used in large networks.
- Hybrid: Combination of two or more topologies.

3.3 Transmission Media

Guided (Wired) Media

- Twisted Pair Cable: UTP (unshielded) and STP (shielded). Used in LANs. Categories: Cat5, Cat5e, Cat6, Cat6a.
- Coaxial Cable: Central conductor with insulating layer and metallic shield. Used for cable TV, older Ethernet.
- Fiber Optic Cable: Uses light signals. Highest bandwidth, longest distance, immune to EMI. Single-mode and Multi-mode.

Unguided (Wireless) Media

- Radio Waves: Omnidirectional. Wi-Fi, AM/FM radio, Bluetooth.
- Microwaves: Unidirectional, line-of-sight. Used in long-distance telecom.
- Infrared: Very short range, line-of-sight. TV remotes, some device-to-device.

3.4 Protocols & Standards

- Ethernet (IEEE 802.3), Wi-Fi (IEEE 802.11), Bluetooth (IEEE 802.15.1), USB.

Physical layer has NO error handling capability. It only transmits bits. Error detection/correction is handled by the Data Link Layer above.

4. Data Link Layer

4.1 Overview

The Data Link Layer (Layer 2) is responsible for node-to-node delivery of data within the same local network. It ensures error-free transmission and hides hardware complexities from upper layers. Data unit = Frame.

Sub-Layers

- LLC (Logical Link Control): Multiplexing, flow control, error messages, acknowledgments.
- MAC (Media Access Control): Device addressing, frame formatting, physical media access control.

4.2 Functions

- Framing: Attaches special bit patterns to beginning/end of data to create frames.
- Physical Addressing: Adds source and destination MAC addresses to frame header.
- Error Detection & Control: Detects and retransmits damaged/lost frames. Uses CRC, Hamming code, Parity bits, Checksum.
- Flow Control: Coordinates data rate between sender and receiver to prevent overflow.
- Access Control: Determines which device has control of the shared channel (CSMA/CD, CSMA/CA, Token Passing).

4.3 Error Detection Techniques

- Parity Bit: Adds 1 bit to make total 1s even (even parity) or odd (odd parity). Detects single-bit errors only.
- Checksum: Sender computes sum of data segments, sends complement. Receiver recomputes to verify.
- CRC (Cyclic Redundancy Check): Uses polynomial division. Stronger than parity, detects burst errors. Most widely used.
- Hamming Code: Can DETECT and CORRECT single-bit errors. Uses redundant bits at power-of-2 positions.

4.4 Error Control Protocols (ARQ)

ARQ (Automatic Repeat reQuest) – retransmission-based error control.

- Stop-and-Wait ARQ: Send one frame, wait for ACK before sending next. Simple but slow. Efficiency = $1/(1+2a)$ where a = propagation delay / transmission time.
- Go-Back-N ARQ: Sender can send N frames (window size) without waiting for ACK. On error, retransmit from errored frame onward. Receiver discards all frames after error.
- Selective Repeat ARQ: Sender can send N frames. On error, only the errored frame is retransmitted. Receiver buffers out-of-order frames. More efficient than GBN but complex.

4.5 Flow Control

- Stop-and-Wait: Send one frame, wait for ACK. Simple but low utilization.

- Sliding Window: Sender maintains a window of frames it can send without ACK. Window slides as ACKs arrive. Used in both GBN and Selective Repeat.

Window Size: Go-Back-N max sender window = $2^n - 1$. Selective Repeat max sender window = 2^{n-1} . Where n = number of bits for sequence number.

4.6 Switching Techniques

- Circuit Switching: Dedicated path established before communication. Used in traditional telephone networks. Guaranteed bandwidth but wasteful if idle.
- Packet Switching: Data broken into packets, each routed independently. Two types:
 - Datagram: Each packet routed independently, may arrive out of order (e.g., IP).
 - Virtual Circuit: Path established before data transfer, all packets follow same route (e.g., ATM, MPLS).
- Message Switching: Entire message sent to intermediate node (store-and-forward). No dedicated path. High delay.

4.7 Protocols & Devices

- Protocols: Ethernet, PPP, HDLC, SDLC, SLIP.
- Devices: Switch, Bridge, NIC, Wireless Access Point, Layer 2 switches.

4.8 VLAN (Virtual LAN)

VLAN logically segments a physical LAN into multiple isolated broadcast domains using managed switches. Devices in different VLANs cannot communicate without a router. Benefits: improved security, reduced broadcast traffic, flexible grouping regardless of physical location.

4.9 Piggybacking

Piggybacking is a technique where the acknowledgment (ACK) for received data is attached to the outgoing data frame instead of sending a separate ACK. This improves efficiency by reducing the number of frames transmitted.

5. Network Layer

5.1 Overview

The Network Layer (Layer 3) handles logical addressing, routing, and end-to-end packet delivery across interconnected networks. Unlike the Data Link Layer (node-to-node within same network), the Network Layer ensures data travels from source host to destination host across multiple networks. Data unit = Packet.

5.2 Key Responsibilities

- Logical Addressing: Assigns unique IP addresses to devices for identification across networks.
- Routing: Determines optimal path for packets using routing algorithms and protocols.
- Packetization: Encapsulates transport layer segments into packets.
- Forwarding: Moves packets from input interface of router to appropriate output interface.
- Fragmentation & Reassembly: Splits large packets to match MTU, reassembles at destination.
- Subnetting: Divides larger networks into smaller subnetworks for efficient addressing.
- NAT (Network Address Translation): Maps private IPs to public IPs, conserving address space.

5.3 IP Addressing

IPv4

- 32-bit address, written in dotted decimal (e.g., 192.168.1.1). Total: $2^{32} = \sim 4.3$ billion addresses.
- Header: 20–60 bytes. Includes version, header length, TTL, protocol, source/destination IP, checksum.

Classful Addressing

Class	Range	Default Subnet Mask	Networks	Hosts/Network
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	126	~16 million
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	~16,000	~65,000
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	~2 million	254
D	224.0.0.0 – 239.255.255.255	Multicast	–	–
E	240.0.0.0 – 255.255.255.255	Reserved/Experimental	–	–

127.0.0.0/8 is reserved for loopback (localhost). 0.0.0.0 is used as default route.

Classless Addressing (CIDR)

CIDR (Classless Inter-Domain Routing) removes fixed class boundaries. Uses prefix notation (e.g., /20) to define network portion. More flexible and efficient allocation of IP space.

IPv6

- 128-bit address – vastly larger address space (2^{128} addresses).
- Written in hexadecimal groups separated by colons (e.g., 2001:0db8::1).
- No NAT needed, built-in IPSec, simplified header (40 bytes fixed), no checksum in header, no fragmentation by routers.

Private vs Public IP

- Private IPs: Used within LANs, not routable on Internet. Ranges: 10.0.0.0/8, 172.16.0.0/12, 192.168.0.0/16.
- Public IPs: Globally unique, assigned by ISP, routable on Internet.

5.4 Subnetting

Subnetting divides a large network into smaller sub-networks (subnets) for better management and security.

- Subnet Mask: Identifies network and host portions. E.g., 255.255.255.0 = /24 → 256 addresses, 254 usable hosts.
- Formula: Number of subnets = 2^s (s = borrowed bits). Hosts per subnet = $2^h - 2$ (h = host bits).
- VLSM (Variable Length Subnet Mask): Allows different subnets to have different sizes. More efficient IP usage.
- Supernetting: Combines multiple smaller networks into a larger one. Opposite of subnetting. Used for route aggregation.

5.5 Routing

Static vs Dynamic Routing

- Static Routing: Routes manually configured by admin. Simple but doesn't adapt to changes. Good for small networks.
- Dynamic Routing: Routes learned automatically via routing protocols. Adapts to topology changes.

Routing Algorithms

- Distance Vector: Each router shares its routing table with neighbors periodically. Uses hop count. E.g., RIP. Suffers from count-to-infinity problem.
- Link State: Each router builds complete network topology map. Uses Dijkstra's algorithm for shortest path. E.g., OSPF, IS-IS. Faster convergence, more complex.
- Path Vector: Used for inter-domain routing. Stores complete path info to prevent loops. E.g., BGP.

Routing vs Flooding

Routing	Flooding
Routing table required	No routing table needed
May give shortest path	Always gives shortest path
Less traffic	More traffic (duplicate packets)
More efficient	More reliable

5.6 Network Layer Protocols

- IP (Internet Protocol): IPv4/IPv6 – provides logical addressing, delivers packets across networks.
- ICMP (Internet Control Message Protocol): Error reporting and diagnostics (ping, traceroute). E.g., destination unreachable.
- ARP (Address Resolution Protocol): Maps IP address to MAC address within a local network.
- RARP (Reverse ARP): Retrieves IP from MAC (obsolete, replaced by DHCP).
- IGMP (Internet Group Management Protocol): Manages multicast group memberships.
- NAT (Network Address Translation): Translates private IPs to public IPs for Internet access.
- IPSec: Secures IP communication through encryption and authentication.
- MPLS (Multiprotocol Label Switching): Uses labels for fast packet forwarding and traffic engineering.
- GRE (Generic Routing Encapsulation): Tunneling protocol that encapsulates packets inside IP packets.

Routing Protocols

- RIP (Routing Information Protocol): Distance vector, hop count metric, max 15 hops, slow convergence.
- OSPF (Open Shortest Path First): Link state, uses Dijkstra, fast convergence, supports VLSM, hierarchical areas.
- IS-IS (Intermediate System to Intermediate System): Link state, used by large ISPs.
- EIGRP (Enhanced Interior Gateway Routing Protocol): Cisco proprietary, hybrid (distance vector + link state features).
- BGP (Border Gateway Protocol): Path vector, used for inter-AS (autonomous system) routing on the Internet. The routing protocol of the Internet.

5.7 DHCP

DHCP (Dynamic Host Configuration Protocol) automatically assigns IP addresses and network configuration (subnet mask, gateway, DNS) to devices. Uses DORA process: Discover → Offer → Request → Acknowledge. Ports 67 (server) and 68 (client).

6. Transport Layer

6.1 Overview

The Transport Layer (Layer 4) provides end-to-end communication between applications on different hosts. It uses port numbers for process-to-process delivery. Implemented only in end systems, not in intermediate routers.

6.2 Functions

- Segmentation & Reassembly: Breaks data into segments (TCP) or datagrams (UDP), reassembles at destination.
- Service Point Addressing: Uses port numbers to deliver data to the correct application process.
- Error Detection & Recovery: Detects errors, retransmits lost/corrupted data (TCP).
- Flow Control: Regulates data rate to prevent receiver overload.
- Multiplexing/Demultiplexing: Multiple applications share one network connection using port numbers.
- Connection Management: Establishes and terminates connections (TCP 3-way handshake).

6.3 TCP (Transmission Control Protocol)

- Connection-oriented, reliable, ordered delivery, error-checking, retransmission.
- Header: 20–60 bytes (variable). Includes source/destination ports, sequence number, ACK number, flags, window size, checksum.
- Use cases: Web browsing (HTTP), email (SMTP), file transfer (FTP).

3-Way Handshake (Connection Establishment)

- Step 1 – SYN: Client sends SYN=1 with its Initial Sequence Number (ISN).
- Step 2 – SYN-ACK: Server replies SYN=1, ACK=1 with its own ISN and ACK = client_ISN + 1.
- Step 3 – ACK: Client sends ACK=1 with ACK = server_ISN + 1. Connection ESTABLISHED.

Why 3 steps and not 2? The third ACK provides mutual confirmation – both sides are sure the other received their SYN. Prevents stale connections.

Connection Termination (4-Way)

- Step 1: Initiator sends FIN. Step 2: Receiver sends ACK.
- Step 3: Receiver sends FIN. Step 4: Initiator sends ACK. Connection closed after TIME_WAIT.

TCP Congestion Control

- Slow Start: Window starts at 1 MSS, doubles each RTT (exponential growth) until threshold (sssthresh).
- Congestion Avoidance: After sssthresh, window increases by 1 MSS per RTT (linear growth).
- Fast Retransmit: On 3 duplicate ACKs, retransmit lost segment immediately (don't wait for timeout).
- Fast Recovery: After fast retransmit, set sssthresh = cwnd/2, cwnd = sssthresh + 3, then linear increase.

6.4 UDP (User Datagram Protocol)

- Connectionless, unreliable, no ordering, no retransmission, minimal overhead.
- Header: Fixed 8 bytes only. Includes source/destination ports, length, checksum.
- Use cases: Live streaming, online gaming, VoIP, DNS queries – speed > reliability.

6.5 TCP vs UDP

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (ACK, retransmit)	Unreliable (no ACK)
Ordering	Ordered delivery	No ordering
Speed	Slower (overhead)	Faster (lightweight)
Header Size	20–60 bytes	8 bytes
Error Checking	Full error checking	Basic checksum only
Flow Control	Yes (sliding window)	No
Use Cases	Web, email, file transfer	Streaming, gaming, VoIP

6.6 Other Transport Protocols

- SCTP (Stream Control Transmission Protocol): Reliable, supports multi-streaming and multi-homing. Used in telecom signaling.
- QUIC: Google's protocol built on UDP. Provides TCP-like reliability with faster connection setup. Used by HTTP/3.
- DCCP (Datagram Congestion Control Protocol): Connectionless with congestion control. For streaming media.
- RUDP (Reliable UDP): Adds reliability features to UDP. Used in some gaming/streaming applications.

7. Session & Presentation Layers

7.1 Session Layer (Layer 5)

Responsible for establishing, managing, and terminating sessions (connections) between applications. Also provides authentication and security.

- Session Establishment, Maintenance, Termination: Creates and manages communication sessions.
- Synchronization: Adds checkpoints in data stream. If failure occurs, retransmission starts from last checkpoint, not beginning.
- Dialog Control: Determines communication mode – half-duplex or full-duplex.
- Protocols: NetBIOS, RPC (Remote Procedure Call), PPTP (Point-to-Point Tunneling Protocol).

7.2 Presentation Layer (Layer 6)

Translation layer – extracts data from application layer and formats it for transmission.

- Translation: Converts between data formats (e.g., ASCII to EBCDIC).
- Encryption/Decryption: Converts plaintext to ciphertext (and back) using keys. E.g., TLS/SSL.
- Compression: Reduces bits for efficient transmission. Especially for multimedia.
- Protocols/Standards: TLS/SSL, MIME, JPEG, MPEG, GIF, AFP (Apple Filing Protocol), NCP.

8. Application Layer

8.1 Overview

The Application Layer (Layer 7) is the highest OSI layer, closest to the end user. It provides network services directly to user applications. In TCP/IP it combines Session + Presentation + Application. Data unit = Message.

8.2 Functions

- Data Representation: Converts user data into standard network-compatible formats (ASCII, HTML, JPEG).
- Network Service Access: Allows apps (email, FTP, browser) to interact with the network.
- Application Protocols: Defines rules for message formatting, data exchange, service requests.
- Session Management: Establishes, manages, terminates communication sessions.

8.3 Key Protocols

DNS (Domain Name System)

- Translates domain names (www.example.com) to IP addresses. Port 53. Uses hierarchical name space.
- Query types: Recursive (resolver does full lookup), Iterative (returns referral). Caching improves speed.
- Records: A (IPv4), AAAA (IPv6), CNAME (alias), MX (mail), NS (name server), PTR (reverse lookup).

HTTP / HTTPS

- HTTP (HyperText Transfer Protocol): Port 80. Stateless request-response protocol for web communication. Methods: GET, POST, PUT, DELETE, PATCH.
- HTTPS: HTTP + TLS/SSL encryption. Port 443. Ensures secure, encrypted web communication.

FTP (File Transfer Protocol)

- Transfers files between client and server. Two connections: Control (Port 21) and Data (Port 20).
- Modes: Active (server connects back to client), Passive (client initiates both connections).

Email Protocols

- SMTP (Simple Mail Transfer Protocol): Sends emails. Ports 25 (unencrypted), 587 (TLS). Push protocol.
- POP3 (Post Office Protocol v3): Downloads emails to client, deletes from server. Port 110. Offline access.
- IMAP (Internet Message Access Protocol): Syncs emails on server, access from multiple devices. Port 143.

Other Protocols

- DHCP: Dynamic IP assignment. Ports 67/68. DORA process.
- SNMP (Simple Network Management Protocol): Monitors and manages network devices. Ports 161/162. Uses agents and managers.
- Telnet: Remote login. Port 23. Unencrypted – replaced by SSH.
- SSH (Secure Shell): Encrypted remote login. Port 22.
- NTP (Network Time Protocol): Synchronizes clocks across networks. Port 123.
- TFTP (Trivial FTP): Simplified FTP using UDP. Port 69. No authentication.
- NFS (Network File System): Remote file access as if local. Port 2049.
- LDAP (Lightweight Directory Access Protocol): Directory services for user/resource lookup. Port 389.
- MQTT: Lightweight messaging for IoT. Publish-subscribe model.

- SIP (Session Initiation Protocol): Sets up, manages, terminates multimedia sessions (VoIP, video calls).

8.4 Client-Server vs Peer-to-Peer

Feature	Client-Server	Peer-to-Peer
Architecture	Centralized server	Decentralized
Scalability	Add more servers	Each peer adds capacity
Management	Easier (centralized)	Harder (distributed)
Security	Better (controlled)	Weaker
Examples	Web, Email, FTP	BitTorrent, Skype

8.5 CDN (Content Delivery Network)

A CDN is a geographically distributed network of proxy servers that cache and deliver content (web pages, videos) from the nearest edge server to the user. Reduces latency, improves load times, and handles traffic spikes. E.g., Cloudflare, Akamai, AWS CloudFront.

8.6 WWW (World Wide Web)

The WWW is a system of interlinked hypertext documents accessed via the Internet using web browsers. Uses HTTP/HTTPS for communication, URLs for addressing, and HTML for content structure. Invented by Tim Berners-Lee in 1989.

9. Network Security

9.1 Overview

Network security protects networks and data from unauthorized access, misuse, and cyberattacks. Ensures confidentiality (only authorized access), integrity (data not tampered), and availability (services accessible when needed) – the CIA triad.

9.2 Security Layers

- Physical Security: Access cards, biometrics, CCTV, locked server rooms. Protects hardware.
- Technical Security: Firewalls, encryption, antivirus, IDS/IPS. Protects data in motion/at rest.
- Administrative Security: Policies, user permissions, authentication, compliance.

9.3 Key Security Mechanisms

Firewalls

- Barrier between trusted and untrusted networks. Controls traffic based on rules.
- Types: Packet filtering (header inspection), Stateful inspection (tracks active connections), Application-level gateway (proxy), Next-gen (deep packet inspection + IPS).

Encryption

- Symmetric: Same key for encrypt/decrypt. Fast. E.g., AES, DES, 3DES.
- Asymmetric: Public key encrypts, private key decrypts. Slower but secure key exchange. E.g., RSA, ECC, Diffie-Hellman.
- Hashing: One-way function producing fixed-size digest. E.g., SHA-256, MD5. Used for integrity verification.

VPN (Virtual Private Network)

- Creates encrypted tunnel over public Internet for secure remote access.
- Protocols: IPSec, SSL/TLS, PPTP, L2TP, OpenVPN, WireGuard.
- Types: Site-to-site (connects networks), Remote access (individual users).

IDS / IPS

- IDS (Intrusion Detection System): Monitors and ALERTS on suspicious activity. Passive.
- IPS (Intrusion Prevention System): Monitors and BLOCKS malicious traffic. Active. Inline deployment.

Authentication

- Something you know (password), something you have (token/card), something you are (biometrics).
- Multi-factor authentication (MFA) combines two or more factors for stronger security.

9.4 Common Network Attacks

- DoS/DDoS: Overwhelms target with traffic, making it unavailable. DDoS uses multiple sources.
- Man-in-the-Middle (MITM): Attacker intercepts communication between two parties.
- Phishing: Social engineering to steal credentials via fake emails/websites.

- ARP Spoofing/Poisoning: Attacker sends fake ARP messages to link their MAC with a victim's IP.
- MAC Spoofing: Changing MAC address to impersonate another device.
- DNS Spoofing: Corrupts DNS cache to redirect users to malicious sites.
- SQL Injection, XSS: Application-layer attacks exploiting input validation flaws.

9.5 Security Protocols

- TLS/SSL: Encrypts data between client and server (HTTPS). TLS is the successor of SSL.
- IPSec: Encrypts and authenticates IP packets. Used in VPNs. Modes: Transport (payload only), Tunnel (entire packet).
- WPA3: Latest Wi-Fi security protocol with stronger encryption.
- SSH: Secure remote access using encrypted tunnels. Replaces Telnet.
- RADIUS / TACACS+: Centralized authentication for network access.

10. Wireless & Modern Networking

10.1 Wireless Networks

Wi-Fi (IEEE 802.11)

- WLAN standard. Versions: 802.11a/b/g/n/ac/ax (Wi-Fi 6). Frequencies: 2.4 GHz and 5 GHz.
- Uses CSMA/CA (Collision Avoidance) for medium access.
- Security: WEP (broken) → WPA → WPA2 (AES) → WPA3 (latest, strongest).

Bluetooth (IEEE 802.15.1)

- Short-range (1–10m typical) wireless. 2.4 GHz. Used for peripherals, audio, IoT.
- Versions: 4.0 (Low Energy), 5.0+ (longer range, higher speed).

Zigbee (IEEE 802.15.4)

- Low-power, low-data-rate wireless for IoT and home automation. Mesh networking, supports thousands of nodes.

Mobile/Cellular Networks

- 1G: Analog voice. 2G (GSM): Digital voice + SMS. 2.5G (GPRS/EDGE): Basic data.
- 3G (UMTS/CDMA2000): Mobile internet, video calls. 4G (LTE): High-speed data, VoIP.
- 5G: Ultra-fast speeds (up to 10 Gbps), ultra-low latency (~1ms), massive device connectivity, network slicing.

10.2 Modern Networking Concepts

SDN (Software-Defined Networking)

- Decouples control plane from data plane. Centralized controller manages network switches/routers programmatically.
- Benefits: Agility, programmability, centralized management, easier policy enforcement.

NFV (Network Functions Virtualization)

- Virtualizes network functions (firewalls, load balancers, routers) on standard servers instead of proprietary hardware.
- Reduces cost, improves scalability, enables rapid deployment.

Cloud Networking

- IaaS (Infrastructure as a Service): Virtual machines, storage, networking. E.g., AWS EC2, Azure VMs.
- PaaS (Platform as a Service): Development platforms. E.g., Heroku, Google App Engine.
- SaaS (Software as a Service): Ready-to-use software. E.g., Gmail, Salesforce, Office 365.

IoT (Internet of Things)

- Network of physical devices embedded with sensors, software, connectivity. Communicate and exchange data over the Internet.
- Protocols: MQTT (lightweight messaging), CoAP (constrained devices), Zigbee, BLE.

Network Slicing (5G)

Divides a single physical 5G network into multiple virtual networks (slices), each optimized for a specific use case (e.g., IoT, autonomous vehicles, mobile broadband). Each slice has its own QoS, security, and capacity settings.

11. Key Comparisons & Formulas

11.1 OSI vs TCP/IP

Feature	OSI Model	TCP/IP Model
Layers	7	4
Developed by	ISO	DARPA / DoD
Nature	Theoretical / Reference	Practical / Implementation
Layer boundaries	Strict	Not rigid
Protocol dependence	Protocol-independent	Protocol-dependent
Usage	Learning & design	Real-world Internet

11.2 Hub vs Switch vs Router

Feature	Hub	Switch	Router
Layer	Physical (1)	Data Link (2)	Network (3)
Address	None	MAC	IP
Data Flow	Broadcast	Unicast	Routed
Intelligence	Dumb	Smart	Smarter
Collision Domain	One (shared)	Separate per port	Separate per port
Broadcast Domain	One	One (default)	Separate per interface

11.3 Circuit vs Packet vs Message Switching

Feature	Circuit	Packet	Message
Path	Dedicated	Dynamic per packet	Store-and-forward
Bandwidth	Reserved (may waste)	Shared efficiently	Shared
Delay	Low (once set up)	Variable	High
Example	Telephone	Internet (IP)	Email (old)

11.4 Important Formulas

Efficiency

- Stop-and-Wait: Efficiency = $1 / (1 + 2a)$, where a = Propagation Delay / Transmission Time.
- Go-Back-N: Efficiency = $N / (1 + 2a)$ if $N < 1+2a$, else 1. Window size $N \leq 2^n - 1$.
- Selective Repeat: Efficiency = $N / (1 + 2a)$ if $N < 1+2a$, else 1. Window size $N \leq 2^{n-1}$.

Transmission & Propagation

- Transmission Delay = Packet Size (L) / Bandwidth (B).
- Propagation Delay = Distance (d) / Speed of signal (v).
- Total Delay = Transmission + Propagation + Queuing + Processing.
- Throughput = Data transferred / Time taken.

Subnetting

- Number of subnets = 2^s (s = bits borrowed from host).
- Hosts per subnet = $2^h - 2$ (h = remaining host bits; subtract 2 for network and broadcast addresses).
- Block size = 256 - subnet mask octet value.

TCP Congestion Window

- Slow Start: cwnd doubles every RTT $\rightarrow$ cwnd = $2^{(\text{round number})}$ MSS, until ssthresh.
- Congestion Avoidance: cwnd += 1 MSS per RTT (linear growth).
- After timeout: ssthresh = cwnd/2, cwnd = 1 MSS (reset to slow start).
- After 3 dup ACKs (Fast Recovery): ssthresh = cwnd/2, cwnd = ssthresh + 3 MSS.

11.5 Important Port Numbers

Port	Protocol	Service
20, 21	FTP	File Transfer (Data, Control)
22	SSH	Secure Shell
23	Telnet	Remote Login (unencrypted)
25, 587	SMTP	Email Sending
53	DNS	Domain Name Resolution
67, 68	DHCP	Dynamic IP Assignment
80	HTTP	Web (unencrypted)
110	POP3	Email Download
143	IMAP	Email Sync
161, 162	SNMP	Network Management
443	HTTPS	Web (encrypted)

12. Interview Questions

Q1. What is a computer network?

Ans: A group of two or more connected devices that communicate to share data, resources, and services using wired or wireless media.

Q2. What is the difference between OSI and TCP/IP models?

Ans: OSI has 7 layers (theoretical reference model by ISO). TCP/IP has 4 layers (practical, protocol-driven, used on the Internet). TCP/IP combines OSI's top 3 layers into Application and bottom 2 into Network Access.

Q3. What is the difference between TCP and UDP?

Ans: TCP is connection-oriented, reliable, ordered, slower (20-60 byte header). UDP is connectionless, unreliable, unordered, faster (8 byte header). TCP for web/email, UDP for streaming/gaming.

Q4. Explain the 3-way handshake.

Ans: TCP connection setup: (1) Client sends SYN with ISN, (2) Server replies SYN-ACK with its ISN and ACK=client_ISN+1, (3) Client sends ACK with ACK=server_ISN+1. Connection established.

Q5. What is a MAC address vs IP address?

Ans: MAC address is a 48-bit physical hardware address (Layer 2, unique to NIC, does not change). IP address is a 32-bit (IPv4) or 128-bit (IPv6) logical address (Layer 3, assigned by network, can change).

Q6. What is ARP?

Ans: Address Resolution Protocol maps an IP address to a MAC address within a local network by broadcasting an ARP request and receiving a unicast ARP reply.

Q7. What is DNS?

Ans: Domain Name System translates human-readable domain names (www.google.com) into IP addresses. Uses hierarchical structure with root, TLD, and authoritative name servers.

Q8. What are the differences between a hub, switch, and router?

Ans: Hub (Layer 1): broadcasts to all ports, dumb. Switch (Layer 2): uses MAC addresses, sends to destination only, smart. Router (Layer 3): uses IP addresses, routes between networks, smarter.

Q9. What is subnetting?

Ans: Dividing a large network into smaller sub-networks for better management, security, and efficient IP usage. Uses subnet masks to define network/host portions. Hosts per subnet = $2^h - 2$.

Q10. What is NAT?

Ans: Network Address Translation maps private IP addresses to public IPs for Internet access. Conserves public IP addresses and adds a layer of security by hiding internal network structure.

Q11. What is a VLAN?

Ans: Virtual LAN logically segments a physical LAN into isolated broadcast domains using managed switches. Improves security, reduces broadcast traffic, enables flexible grouping.

Q12. Explain DHCP and its process.

Ans: Dynamic Host Configuration Protocol automatically assigns IP addresses. DORA: Discover (client broadcasts), Offer (server offers IP), Request (client accepts), Acknowledge (server confirms).

Q13. What is a firewall?

Ans: A security device/software that monitors and controls network traffic based on predefined rules. Types: packet filtering, stateful inspection, application-level gateway, next-gen firewall.

Q14. What is the difference between IDS and IPS?

Ans: IDS (Intrusion Detection) monitors and alerts on suspicious activity (passive). IPS (Intrusion Prevention) monitors and blocks malicious traffic (active, inline).

Q15. What is a VPN?

Ans: Virtual Private Network creates an encrypted tunnel over the public Internet for secure remote access. Uses protocols like IPSec, SSL/TLS, WireGuard.

Q16. What is ICMP used for?

Ans: Internet Control Message Protocol is used for error reporting and diagnostics. E.g., ping (echo request/reply) and traceroute (destination unreachable, TTL exceeded).

Q17. What is the difference between IPv4 and IPv6?

Ans: IPv4: 32-bit, ~4.3B addresses, NAT required, variable header (20-60 bytes), checksum in header. IPv6: 128-bit, vastly larger, no NAT needed, fixed 40-byte header, no checksum, built-in IPSec.

Q18. What is congestion control in TCP?

Ans: Mechanisms to prevent network congestion: Slow Start (exponential growth to ssthresh), Congestion Avoidance (linear growth), Fast Retransmit (on 3 dup ACKs), Fast Recovery (halve window).

Q19. What is the difference between flow control and congestion control?

Ans: Flow control prevents receiver overflow (between sender-receiver pair using sliding window). Congestion control prevents network overload (adjusts sending rate based on network conditions).

Q20. What are the different types of network topologies?

Ans: Bus (shared backbone), Star (central hub), Ring (circular), Mesh (all-to-all), Tree (hierarchical), Hybrid (combination). Mesh is most reliable; Star is most common.

Q21. What is CSMA/CD and CSMA/CA?

Ans: CSMA/CD (Collision Detection): Used in wired Ethernet. Detects collision, stops, waits random time, retransmits. CSMA/CA (Collision Avoidance): Used in Wi-Fi. Avoids collisions using RTS/CTS mechanism.

Q22. What is the sliding window protocol?

Ans: Flow control mechanism where sender maintains a window of frames that can be sent without ACK. Window slides as ACKs arrive. Used in Go-Back-N (window $\leq 2^n - 1$) and Selective Repeat (window $\leq 2^n - 1$).

Q23. What is HTTP vs HTTPS?

Ans: HTTP (Port 80) transfers data in plaintext. HTTPS (Port 443) adds TLS/SSL encryption for secure communication. HTTPS is standard for all modern websites.

Q24. What is a socket?

Ans: A socket is an endpoint for communication, defined by an IP address + port number pair (e.g., 192.168.1.1:8080). Used by TCP/UDP for process-to-process communication.

Q25. What is the difference between unicast, broadcast, and multicast?

Ans: Unicast: one-to-one (single sender to single receiver). Broadcast: one-to-all (sent to all devices in network). Multicast: one-to-many (sent to a specific group of receivers).

Q26. Explain the concept of port numbers.

Ans: Port numbers (0-65535) identify specific processes/applications. Well-known: 0-1023 (HTTP=80, SSH=22). Registered: 1024-49151. Dynamic/Private: 49152-65535.

Q27. What is SDN?

Ans: Software-Defined Networking decouples the control plane from data plane. A centralized controller manages network devices programmatically, enabling agility and easier management.

Q28. What is QoS?

Ans: Quality of Service refers to mechanisms that prioritize certain types of network traffic (e.g., video calls over file downloads) to ensure performance. Techniques: traffic shaping (Token Bucket, Leaky Bucket), priority queuing.

Q29. What are Token Bucket and Leaky Bucket algorithms?

Ans: Leaky Bucket: Smooths bursty traffic to a constant output rate (bucket leaks at fixed rate). Token Bucket: Allows controlled bursts – tokens added at fixed rate, each packet needs a token to transmit. More flexible than Leaky Bucket.

Q30. What is network slicing in 5G?

Ans: Divides a single physical 5G network into multiple virtual networks (slices), each tailored for specific use cases (IoT, autonomous vehicles, broadband) with its own QoS, security, and capacity.